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Day-ahead Numerical Weather Prediction solar irradiance correction using a clustering method based on weather conditions

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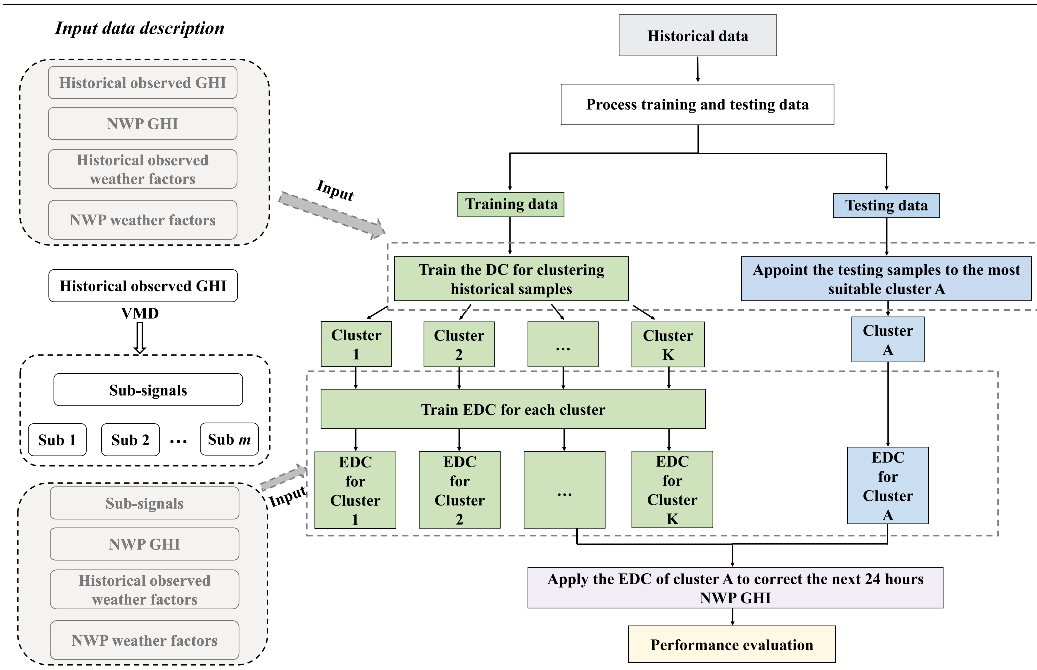
GRAPHICAL ABSTRACT



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ABSTRACT

Accurate solar irradiance forecasts can make solar power forecasts more reliable, which can help the power grid dispatch reasonably. In practice, Numerical Weather Prediction (NWP) is widely applied in solar irradiance forecasts. In this paper, the statistical NWP Global Horizontal Irradiance (GHI) error analysis shows that the characteristics of NWP GHI error vary obviously under different weather conditions. However, existing correction methods are not designed contrapuntally for different weather conditions, resulting in poor

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Clustering
Variational mode decomposition

correction performance. To solve this problem, a hybrid method is proposed to get day-ahead correction results for NWP GHI. Specifically, the hybrid method consists of three key parts, including Deep Clustering (DC), Variational Mode Decomposition (VMD), and an Encoder–Decoder based Correction model (EDC). DC is used to categorize the historical samples into three clusters, the input of which is the multi-dimensional information series containing observed data and NWP data. After a feature selection by VMD, the correction models are trained on each cluster respectively. The performance of the proposed model is evaluated with a public dataset and an actual field dataset, and the results demonstrate that the accuracy has been effectively improved by the proposed method compared with other models. In addition, we find that adopting VMD is effective in improving the correction accuracy, and the root mean square error is reduced by 5.70% and 9.32% compared with models without it.

Nomenclature

γ	Coefficient balancing clustering loss and reconstruction loss
λ^1	Learning rate
μ_j	Cluster center
$f_{\text{en}}(\cdot)$	Encoder
f_i	NWP GHI at the moment i
$g_{\text{de}}(\cdot)$	Decoder
GHI_{cs}	Clear-sky GHI
$I^{original}$	Original information data
K_{cs}	Clear sky index
$K_{cs}(\text{day})$	Daily clear sky index
L_{cl}	Clustering loss
L_{re}	Reconstruction loss
O^n	Observed weather factors
o_a	The average actual GHI observations
o_i	Actual GHI observations at the moment i
P	Auxiliary target distribution
P^n	NWP weather factors
Q	Student's t-distribution
S^{de}	The input of decoder
S^{en}	The input of encoder
W_{de}	The parameter of decoder
W_{en}	The parameter of encoder
X^O	Observed GHI
X^P	NWP GHI
Z	Latent representation

Acronyms

AnEn	Analog ensemble
ANNs	Artificial neural networks
AT	Ambient temperature
BPNN	Back propagation neural network
CLSTM	Convolutional long short-term memory
CNN	Convolutional neural network
DC	Deep clustering
ECMWF	European Centre for Medium-Range Weather Forecasts
EDC	Encoder–decoder based correction model
EEMD	Ensemble empirical mode decomposition
EMD	Empirical mode decomposition
FC	Fully connected layer
FCM	Fuzzy C-Means
GBR	Gradient boosted regression
GHI	Global horizontal irradiance
GRU	Gated recurrent unit
LSTM	Long short-term memory
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MLP	Multiple layer perceptron
MOS	Model output statistics
nMAE	Normalized mean absolute error
nRMSE	Normalized root mean squared error
NWP	Numerical Weather Prediction
PV	Photovoltaic
RFR	Random forest regression
RH	Relative humidity
RI	Rand index
RMSE	Root mean squared error
SVR	Support vector regression
TFT	Temporal fusion transformer
VMD	Variational mode decomposition
WD	Wavelet decomposition
WRF	Weather Research and Forecasting
XGBoost	Extreme gradient boosting

1. Introduction

Solar energy is one of the most popular forms of renewable energy, offering advantages such as abundant resources, free usage, no need for transportation, and minimal pollution [1,2]. As solar power becomes more widely utilized, the daily operation and management of power grids, which includes ensuring power system reliability and managing voltage fluctuations, becomes more complex due to the significant variability and unschedulability of solar electricity generation. Accurate forecasting of solar power production is beneficial for the operation and management of power systems, including solar power plants [3–6]. Since solar power production is heavily dependent on solar irradiance [7], the solar power output forecast can be transformed into solar irradiance forecast [8].

There are several approaches to forecasting solar irradiance, including Numerical Weather Prediction (NWP) models, persistence models, and cloud trajectory models [9]. NWP is more commonly employed for forecasts extending beyond 6 hours, based on the forecast horizon [10]. However, NWP models are prone to forecast errors due to uncertainties in model input, parameter estimation, and model structure [11]. Previous studies have developed various post-processing approaches to improve NWP model output, including Model Output Statistics (MOS), analog ensemble (AnEn), and machine learning. MOS utilizes multivariate linear regression with the predictors of model prediction variables and prior observations to develop more accurate predictions at specified locations [12]. On the other hand, AnEn searches for previous historical forecasts that are similar to the current model forecasts and then uses the corresponding observed values of these historical forecasts to correct the current forecasts [13].

Recently, machine learning methods have gained increasing attention and have been employed as post-processing correction models to enhance the accuracy of the NWP Global Horizontal Irradiance (GHI) output, compared to other two methods. The machine learning approach offers a more precise and informative alternative to data modeling for large complex datasets. In terms of NWP output correction, numerous machine learning methods have been proposed in the literature. For instance, Support Vector Regression (SVR)-based models have been utilized for regional GHI forecast of specific NWP models, demonstrating significant performance improvements achievable through machine learning methods [14]. Additionally, the adoption of Artificial Neural Networks (ANNs) has led to notable enhancements in refining short-term solar irradiance forecasts from the Weather Research and Forecasting (WRF) outputs [15]. Furthermore, Convolutional Neural Network (CNN) has been employed as a post-processing technique to correct the WRF one-day simulation outputs [16]. The Long Short-Term Memory (LSTM) has also been harnessed to produce NWP wind-power correction results [17]. Moreover, hybrid models combining two or more machine learning approaches have been developed to improve the precision of NWP output correction. For instance, a hybrid method that combines SVR, Gradient Boosted Regression (GBR), and Random Forest Regression (RFR) was utilized to downscale and enhance the accumulated radiation forecasts generated by the NWP system in Spain [18].

The correction results are sensitive to changes in weather conditions, presenting a challenge for a single model to generate accurate NWP correction results under varying weather conditions. To address this issue, clustering has been commonly employed as a method to obtain clusters with different characteristics [19]. The combination of clustering methods and regression has been developed to obtain relevant forecast models for different weather conditions. For example, in one study, a K-means clustering approach was utilized to group meteorological data into three clusters, and SVR models were trained for each cluster to obtain one-hour ahead forecasts of solar irradiance [20]. Azimi et al. proposed the TB_K-means method to cluster solar irradiance patterns, employing the Multiple Layer Perceptron (MLP) for hourly solar irradiance forecasts [21]. In another study, daily GHI forecasts were categorized by K-means, and the forecast outputs were corrected for each cluster [10]. Traditional clustering methods are commonly adopted in the majority of current hybrid approaches that use clustering methods for solar forecast [19,21–23]. However, these methods may yield sub-optimal clustering results because they do not jointly consider feature extraction and clustering. Recognizing this limitation, a clustering method based on deep learning was proposed, which used an encoder–decoder framework to learn the latent representation of GHI time sequences for clustering. In this method, the encoder was a Gated Recurrent Unit (GRU) model and the decoder was an MLP model [24]. Also, CNN was used to robustly extract features from input information [25,26].

However, the above methods have the following limitations. Firstly, the majority of existing NWP GHI correction models do not consider separate modeling for different weather conditions, yet the NWP GHI error varies significantly depending on the weather conditions. Secondly, as data becomes more complex, traditional clustering methods lose their advantages in dealing with multi-dimensional data. Additionally, traditional clustering methods make assumptions, such as representing data as vectorized features through various representation learning techniques [10,20,21]. Furthermore, in traditional clustering methods, the clustering and representation learning are conducted separately, rather than mutually reinforcing each other, which may result in sub-optimal clustering results. Thirdly, the input of the aforementioned post-processing correction methods [14,15] is GHI sequences, which lack feature selection and may lead to redundant information input. Similarly, in the clustering process described in [24], the input is GHI sequences, and inadequate extraction of effective features in the clustering process may result in poor performance. In contrast, the input of Deep Clustering (DC) extends to multi-dimensional data, which contains more meaningful information.

In this study, an analysis of NWP GHI error has been conducted, revealing variations in NWP GHI error under different weather conditions. A hybrid method based on deep learning has been proposed to correct NWP GHI for different weather conditions. This method consists of DC, Variational Mode Decomposition (VMD), and an Encoder–Decoder based Correction model (EDC). DC categorizes the information series (including observed data and NWP data) with similar patterns into the same clusters using highly informative features mined by Convolutional Long Short-Term Memory (CLSTM). This approach enables collective learning of feature extraction and clustering. Subsequently, a corresponding day-ahead NWP GHI correction model is trained with the data from the respective cluster. For time-series data, filter methods such as Empirical Mode Decomposition (EMD) [27], Ensemble Empirical Mode Decomposition (EEMD) [28], Wavelet Decomposition (WD) [29], and VMD [30] have been widely applied for feature selection. These methods aim to exploit the high-level useful features of GHI series to enhance forecasting ability. Among these methods, VMD has been found to be particularly promising compared to EEMD and WD, as it does not require heavy computation and demonstrates good robustness [31,32]. Therefore, VMD is utilized to extract features from historical GHI data. The multi-dimensional data, comprising the extracted features and other relevant data in each cluster, are then used as input to train the correction model, which is an encoder–decoder framework designed to produce day-ahead NWP GHI correction results. In this framework, the encoder and decoder are implemented using CLSTM, with CNN being employed to robustly extract features from the input information [25,26]. The major contributions of this work are presented as follows:

(1) NWP GHI error analysis shows the significant variations in NWP GHI error under different weather conditions, providing a foundation for enhancing correction accuracy.
(2) DC is introduced to generate multiple clusters under varying weather conditions. This method categorizes information series (including observed data and NWP data) with similar patterns into the same cluster, optimizing both feature learning and clustering assignments.
(3) EDC is developed for day-ahead NWP GHI correction of each cluster categorized by DC, where the encoder and decoder are the CLSTM. In both the clustering and regression steps, CLSTM is utilized to learn effective features from multi-dimensional input data.
(4) VMD is employed to remove redundant information from historical observed GHI data, thereby contributing to the extraction of meaningful features to enhance the correction results.

The rest of the paper is organized as follows: In Section 2, the NWP GHI error analysis is carried out to provide a basis for correction work. The methodology for day-ahead NWP GHI correction model used in this study is described in Section 3. The results and discussion are presented in Section 4. Several evaluation indexes are used to compare the performance of the proposed hybrid correction model with NWP, Back Propagation Neural Network (BPNN), eXtreme Gradient Boosting (XGBoost), SVR, LSTM, Temporal Fusion Transformer (TFT). Finally, conclusions are drawn in Section 5.

2. NWP GHI error analysis

2.1. Data description

The data used in this work are from two sources. The first dataset (solar plant 1) is a publicly available dataset from a solar plant in Australia, consisting of observed GHI, observed Ambient Temperature (AT), observed Relative Humidity (RH), NWP GHI, NWP AT and NWP RH. And the second dataset (solar plant 2) is an actual field dataset consisting of observed GHI, observed AT, NWP GHI and NWP AT, which is from a solar plant located in certain Province of China. In this work,

Table 1
Details of the datasets used in this study.

Station	NWP model	Period	Temporal resolution	Meteorological information
Solar plant 1	ECMWF	01/01/2019–31/12/2022	1 (h)	GHI (W/m^2) AT (°C) RH (%)
Solar plant 2	WRF	01/06/2018–31/07/2019	15 (min)	GHI (W/m^2) AT (°C)

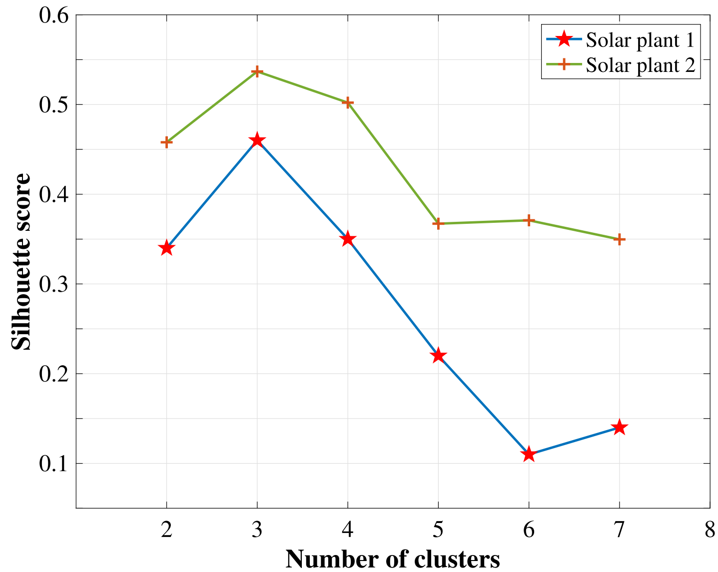


Fig. 1. The Silhouette score under different number of clusters.

the NWP forecasts of solar plant 1 and solar plant 2 are from the European Centre for Medium-Range Weather Forecasts (ECMWF) and the WRF model for mesoscale weather systems developed for emulation by the United States respectively. The forecasts start at 0:00 every day and run up to 24 hours ahead (day-ahead). Details of the solar plants and datasets used in this study are shown in Table 1.

2.2. Weather condition definition

Previous works [33] have been carried out based on three weather conditions (sunny, overcast and cloudy). For our datasets, using a clustering algorithm to divide the samples of the dataset into several clusters. The optimal number of clusters is measured by silhouette score, a kind of measurement to evaluate the clustering approaches performance [24]. From Fig. 1, the optimal number of clusters for solar plant 1 and solar plant 2 are both 3, which is consistent with existing work. Therefore, we consider dividing three weather conditions, including sunny, overcast and cloudy.

Generally, the clear sky index (K_{cs}) is used to characterize different weather conditions [6,34], and it is calculated by:

$$K_{cs} = GHI/GHI_{cs} \tag{1}$$

K_{cs}(day) is the daily clear sky index:

$$K_{cs}(\text{ day }) = \frac{\sum_{i=1}^{n} GHI(i)}{\sum_{i=1}^{n} GHI_{cs}(i)} \tag{2}$$

where n denotes the amount of the samples in one day. Specifically, for solar plant 1 and solar plant 2, n is 24 and 96, respectively. In our work, sunny conditions are defined by $K_{cs}(\text{day}) > 0.9$, cloudy conditions by $0.5 < K_{cs}(\text{day}) < 0.9$, and overcast conditions by $K_{cs}(\text{day}) < 0.5$. Moreover, the threshold values of K_{cs}(day) are advised to tune on the specific solar plant [35].

Table 2
Statistical description of NWP GHI error for different weather conditions (W/m^2).

Station	Weather conditions	Positive error		Negative error	
		Average	Maximum	Average	Maximum
Solar plant 1	Sunny	28.30	277.91	−77.44	−762.86
	Overcast	130.25	643.04	−70.51	−416.17
	Cloudy	101.02	766.99	−100.95	−519.74
Solar plant 2	Sunny	81.40	713.52	−124.65	−540.15
	Overcast	271.72	935.16	−82.70	−581.45
	Cloudy	165.06	998.32	−128.66	−955.15

2.3. Error analysis

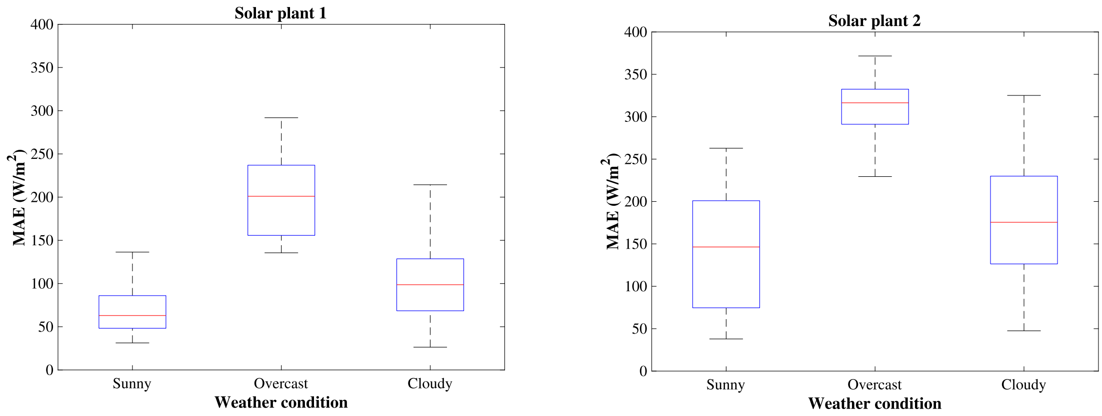
In actual engineering projects, there are many evaluation criteria to measure the forecast performance of the NWP model, but more attention is paid to the deviation between the observed values and the forecast values. That is, two evaluation criteria, NWP GHI error and Mean Absolute Error (MAE). The datasets presented above are taken to analyze the NWP GHI error characteristics under different weather conditions. The NWP GHI error and MAE are calculated as

$$\text{error} = f_i - o_i \tag{3}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |f_i - o_i| \tag{4}$$

where f_i and o_i are the NWP GHI and actual GHI observations at the moment i, respectively. n represents the total number of samples.

The statistics of NWP GHI error in different weather conditions are listed in Table 2. The average positive NWP GHI error in sunny conditions is the smallest among three weather conditions, while it is largest in overcast days. In terms of cloudy conditions, the NWP GHI error in solar plant 1 and solar plant 2 range from −519.74 W/m^2 to 766.99 W/m^2 and −955.15 W/m^2 to 998.32 W/m^2, which are widest



(a) (b)

Fig. 2. Daily MAE statistics under different weather conditions. (a) Solar plant 1, (b) Solar plant 2.

among three weather conditions. The daily MAE between NWP GHI values and observed GHI values under sunny, overcast and cloudy weather conditions is calculated, as shown in Fig. 2. The daily MAE is generally larger in overcast conditions than in sunny and cloudy conditions. And the range of daily MAE on cloudy days is wider than on sunny days. In summary, the NWP GHI error varies obviously under different weather conditions. Therefore, NWP GHI should be corrected for different weather conditions.

3. Methodology

In order to correct the NWP GHI more effectively, the NWP GHI error under different weather conditions is analyzed. Based on the analysis results, this study proposes a hybrid method for day-ahead NWP GHI correction, which comprises two parts: DC and EDC. The steps are as follows:

(1) Framework of day-ahead NWP GHI correction method is shown in Section 3.1.
(2) DC is introduced for categorizing the information data of historical samples and EDC is proposed for day-ahead NWP GHI correction. The details are described in Sections 3.2 and 3.3.
(3) Evaluation criteria are shown in Section 3.4.

3.1. Day-ahead NWP GHI correction framework

This section describes the construction of the hybrid method for day-ahead NWP GHI correction. Based on the results of the Section 2, it can be concluded that NWP GHI should be corrected for different weather conditions. Therefore, DC is applied to classify historical samples under different weather types more accurately. Then, the EDC is developed for day-ahead NWP GHI correction of each cluster categorized by DC. Fig. 3 shows the framework of the proposed hybrid method. Data descriptions of two solar plants are shown in Table 1. The weather factors of solar plant 1 are AT and RH, while there is only AT in solar plant 2. Each dataset is divided into the training set and the testing set, respectively. The training set for solar plant 1 is from Jan. 2019 to Dec. 2021 and the testing set is from Jan. 2022 to Dec. 2022. For solar plant 2, data of Jul. 2019 are extracted as testing set and the remainder as training set.

Firstly, DC is used to obtain multiple clusters under different weather conditions, so that the correction task can be divided into multiple sub-tasks under different weather conditions. At the DC stage, the input contains historical observed GHI data, historical observed weather factors data, NWP GHI data and NWP weather factors data. It should be noted that the input observed data is before the correction time and the purpose of fusing the observed data is to make the clustering results more reliable. To obtain more meaningful features, VMD is used to decompose the historical observed GHI series. For the decomposition process, a time-window is fixed and slides in each decomposition to ensure that the sub-signals are obtained based on the historical data before each correction time. The decomposition results together with the historical observed weather factors data, NWP GHI data and NWP weather factors data are used as the input of EDC to obtain the final results of day-ahead NWP solar irradiance correction.

3.2. Deep clustering (DC)

The DC learns latent representation Z of original information data $I^{original}$. To better classify historical samples based on weather conditions, we adopt observed GHI, observed weather factors, NWP GHI and NWP weather factors as input. The observed data are entirely historical data before the correction time t. And the length of each features (both observed data and NWP data) is one day. Fig. 4 shows the overview of DC.

The CLSTM [36] model as the encoder maps the original information sequences into latent representation and Fully Connected layer (FC) as the decoder reconstruct latent representation back to information sequences. The CLSTM parameters were configured as seen in Table 3. It is worth mentioning that in this work, the ability of the decoder is inferior to that of the encoder, thus the encoder is compelled to effectively learn the potential representation, rather than highly dependent on the decoder [24].

Clustering is performed on latent representation space. There are two key loss functions, namely, reconstruction loss and clustering loss. The clustering center and latent representation of original data are updated by means of combining clustering loss and reconstruction loss after initial clustering centers in latent representation space have been given. Data in the latent representation space is forced to go closer and closer to corresponding clustering center by clustering loss. The reconstruction loss is used to maintain the internal structure of data and avoid the distortion of latent representation space.

For providing a great initialization, the DC is first trained with the reconstruction loss. The reconstruction loss is calculated by:

$$L_{\mathrm{re}} = \left\| I - g_{\mathrm{de}}\left(f_{\mathrm{en}}\left(I^{original}\right)\right)\right\|_2^2 \tag{5}$$

where I indicates original information data, $f_{\mathrm{en}}(\cdot)$ denotes encoder (CLSTM model), and $g_{\mathrm{de}}(\cdot)$ indicates decoder (FC model). Then, derive initial cluster centers using the K-means clustering on the latent representation space. Finally, fine-tune the DC model with clustering loss and reconstruction loss collectively. The clustering loss can be expressed:

$$L_{\mathrm{cl}} = KL(P\|Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \tag{6}$$

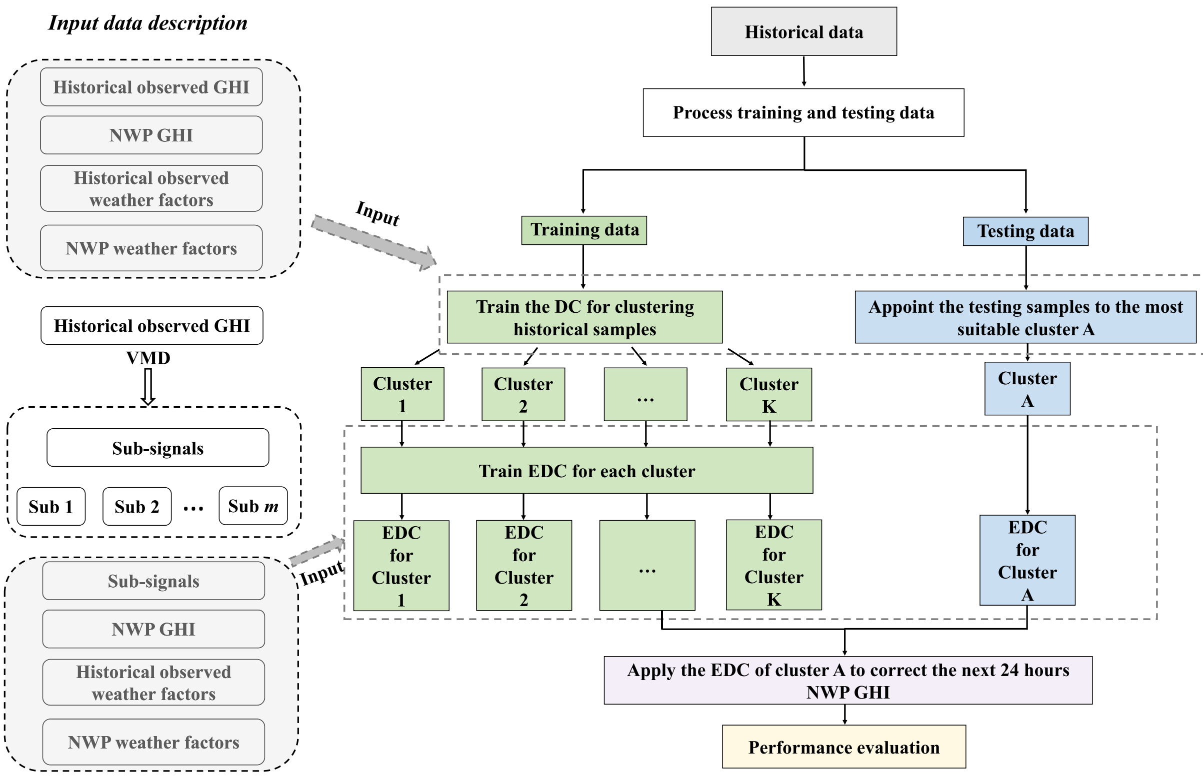


Fig. 3. Flowchart of the proposed model.

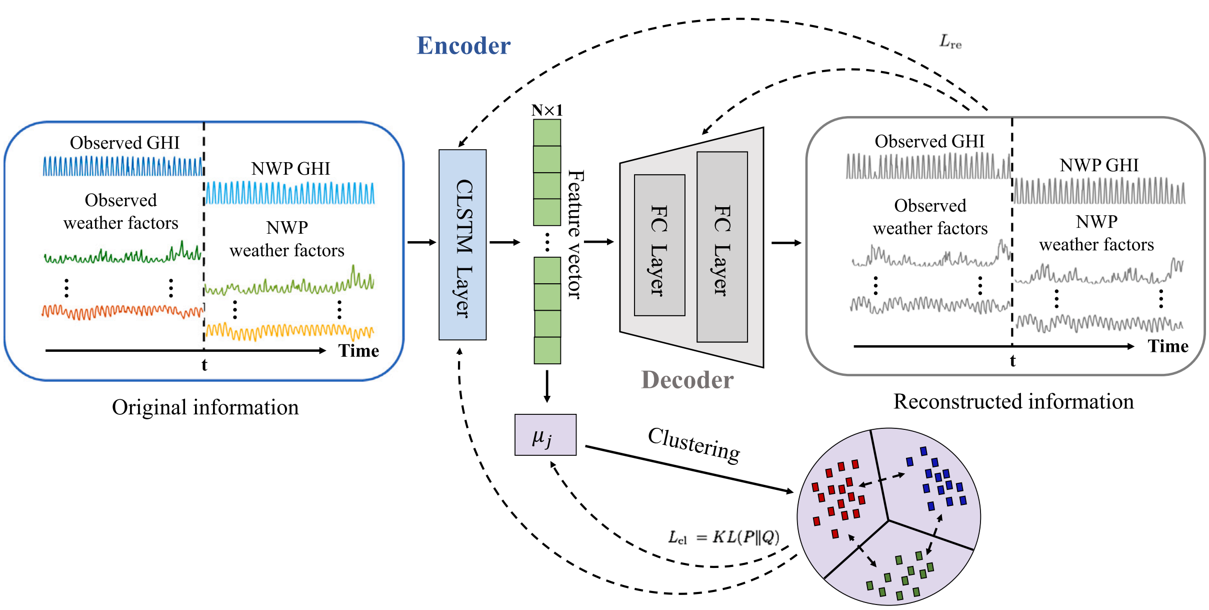


Fig. 4. The overview of DC. The input observed data are historical data before the correction time t.

$$q_{ij} = \frac{\left(1 + \left\|z_i - \mu_j\right\|^2\right)^{-1}}{\sum_j \left(1 + \left\|z_i - \mu_j\right\|^2\right)^{-1}} \tag{7}$$

$$p_{ij} = \frac{q_{ij}^2 / \sum_i q_{ij}}{\sum_j \left(q_{ij}^2 / \sum_i q_{ij}\right)} \tag{8}$$

where $KL(P\|Q)$ indicates Kullback–Leibler divergence, P denotes auxiliary target distribution and Q denotes Student's t-distribution. q_{ij} denotes probability of assigning sample i to cluster j, z_i denotes latent representation data point and μ_j denotes cluster center.

The parameters of the DC model including the parameters of encoder, the parameters of decoder and the cluster center are updated according to the equations as below:

$$\mu_j = \mu_j - \frac{\lambda^1}{n} \sum_{i=1}^{n} \frac{\partial L_{cl}}{\partial \mu_j} \tag{9}$$

$$W_{de} = W_{de} - \frac{\lambda^1}{n} \sum_{i=1}^{n} \frac{\partial L_{re}}{\partial W_{de}} \tag{10}$$

Table 3
Configuration of the various layers of the CLSTM model.

Proposed model		
Conv1D	Filters Kernel size Activation	64 3 Relu
Conv1D	Filters Kernel size Activation	64 3 Relu
MaxPooling	Pool size	2
Flatten	–	–
Repeat vector	–	–
LSTM	Hidden node activation Return sequence	128 Relu True
Time Distributed	Dense Activation	128 Relu
Time distributed	Dense	1
Output (solar plant 1)	–	24
Output (solar plant 2)	–	96

$$W_{\mathrm{en}} = W_{\mathrm{en}} - \frac{\lambda^1}{n}\sum_{i=1}^{n}\left(\frac{\partial L_{\mathrm{re}}}{\partial W_{\mathrm{en}}} + \gamma\frac{\partial L_{\mathrm{cl}}}{\partial W_{\mathrm{en}}}\right) \tag{11}$$

where λ^1 is the learning rate and n denotes the amount of the samples. W_{en} and W_{de} denote the parameter of encoder and that of decoder respectively. γ denotes coefficient balancing clustering loss and reconstruction loss.

3.3. Encoder–decoder based correction model (EDC)

The EDC method is based on an encoder–decoder framework. The input of encoder and decoder is S^{en} and S^{de} represently. It is assumed that every single day is seen as a data object consisting of observed GHI (X^O), observed weather factors (O^n), NWP GHI (X^P) and NWP weather factors (P^n). The VMD [37] algorithm is used for decomposing historical observed GHI series signal into various intrinsic mode functions. Hilbert transform is applied for calculating analytic signal in estimating mode bandwidth. It removes redundant information and obtains effective features of GHI by VMD algorithm. Therefore it helps improve correction preciseness while utilizing effective features as the inputs of correction model. In this work, we fix a time-window with a length of one year, and make the time-window slide in each decomposition to obtain new sub-signals while also preventing information leakage. The number of sub-signals in this paper is 7.

As illustrated in Fig. 5, l and h denote the length of input of encoder and decoder. In this work, the input of the encoder is the seven-day data before the correction time, and the input of the decoder is the one-day data after the correction time. The details are shown in Fig. 5 including NWP GHI, other NWP and observed weather factors, and the features of historical observed GHI obtained by VMD.

It can be seen from Fig. 6 that the encoder and decoder are both CLSTM. For encoder, the normalized input data S^{en} is introduced into CNN model to output a new feature vector. The new feature vector and the h_{k-1} output by the LSTM unit at time $k-1$ are introduced into the LSTM unit at time k, and then continuously circulated. At time t, c_t and h_t are obtained as the final encoding results into the decoder. For decoder, c_t and h_t are the initial state of the first LSTM unit. At time k, the input of decoder is X_k^P, P_k^n and $\hat{y}_{k-1}$, where $\hat{y}_{k-1}$ is the correction result at time $k-1$. $\hat{y}_k$ obtained after CNN, LSTM and FC, c_k and h_k which are the output of LSTM unit at time k are introduced into the time step $k+1$, and then continuously circulated. Finally, the correction results are obtained by decoding. The specific configurations are the same as in the clustering process, as detailed in Table 3.

Table 4
The day-ahead NWP GHI correction performance of proposed approach.

Station	Models	Evaluation criteria		
		RMSE (W/m²)	MAE (W/m²)	nRMSE (%)
Solar plant 1	NWP	109.08	50.47	42.51
	EDC	90.75	35.40	35.36
	DC-EDC	86.61	38.73	33.75
	VMD-EDC	84.53	38.15	32.94
	Hybrid	**75.51**	**34.21**	**29.43**
Solar plant 2	NWP	208.97	111.08	93.24
	EDC	157.83	79.31	70.42
	DC-EDC	138.07	67.63	61.61
	VMD-EDC	138.36	72.78	61.74
	Hybrid	**120.64**	**62.56**	**53.83**

3.4. Evaluation criteria

Evaluation indexes adopt Root Mean Squared Error (RMSE), MAE, normalized Root Mean Squared Error (nRMSE), normalized Mean Absolute Error (nMAE) and Mean Absolute Percentage Error (MAPE) while evaluating the performance of the proposed NWP GHI correction model:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(f_i - o_i\right)^2} \tag{12}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|f_i - o_i\right| \tag{13}$$

$$\mathrm{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(f_i - o_i\right)^2}}{o_a} \times 100\% \tag{14}$$

$$\mathrm{nMAE} = \frac{\frac{1}{n}\sum_{i=1}^{n}\left|f_i - o_i\right|}{o_a} \times 100\% \tag{15}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{f_i - o_i}{o_i}\right| \times 100\% \tag{16}$$

where f_i and o_i denote the NWP GHI and actual GHI observations at the moment i, respectively. n denotes total number of samples. o_a denotes the average actual GHI observations.

4. Results and discussion

To verify the reasonableness and performance of the proposed hybrid method, a series of experiments are carried out in this section.

4.1. Ablation study

The proposed NWP GHI correction model mainly includes two parts, DC and VMD-EDC. An ablation study was conducted to validate the effectiveness of each part. From Table 4, the results in bold black mean the best performance. According to the above metrics, VMD-EDC outperforms EDC, showing the necessity of characteristic selection of GHI. VMD can mine the features of the observed GHI in each cluster to remove some redundant information in the GHI sequence, thus VMD-EDC can get better performance compared with EDC. DC is applied to categorize the historical samples with similar patterns into the same cluster for training several correction models under different weather conditions. For the sample to be corrected with unknown weather conditions, we first appoint it to the most suitable cluster, and then apply the correction model trained by the historical samples in this cluster to obtain its correction results. DC-EDC has a better performance compared with EDC, which shows the superiority of introducing DC. Furthermore, the performance of method Hybrid is best among EDC, DC-EDC and VMD-EDC, which shows it is beneficial for improving the accuracy of NWP GHI correction performance to combine the EDC with the DC and VMD.

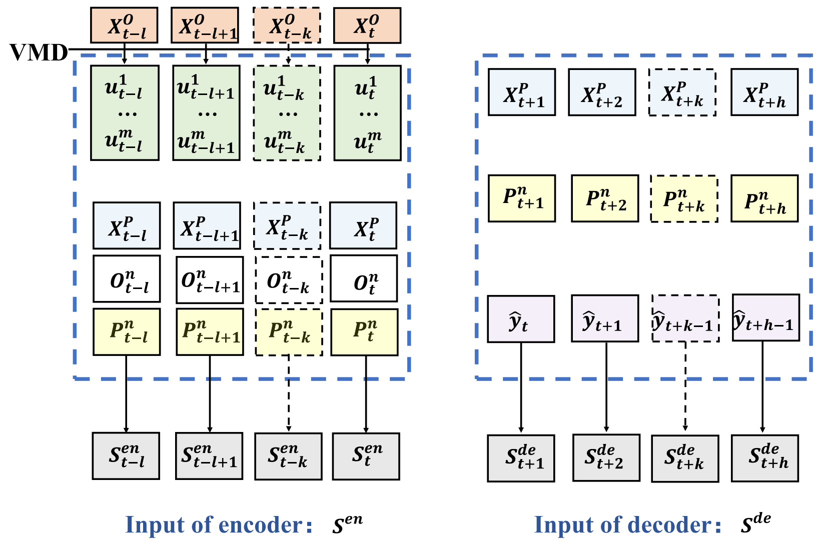


Fig. 5. The input of encoder and decoder.

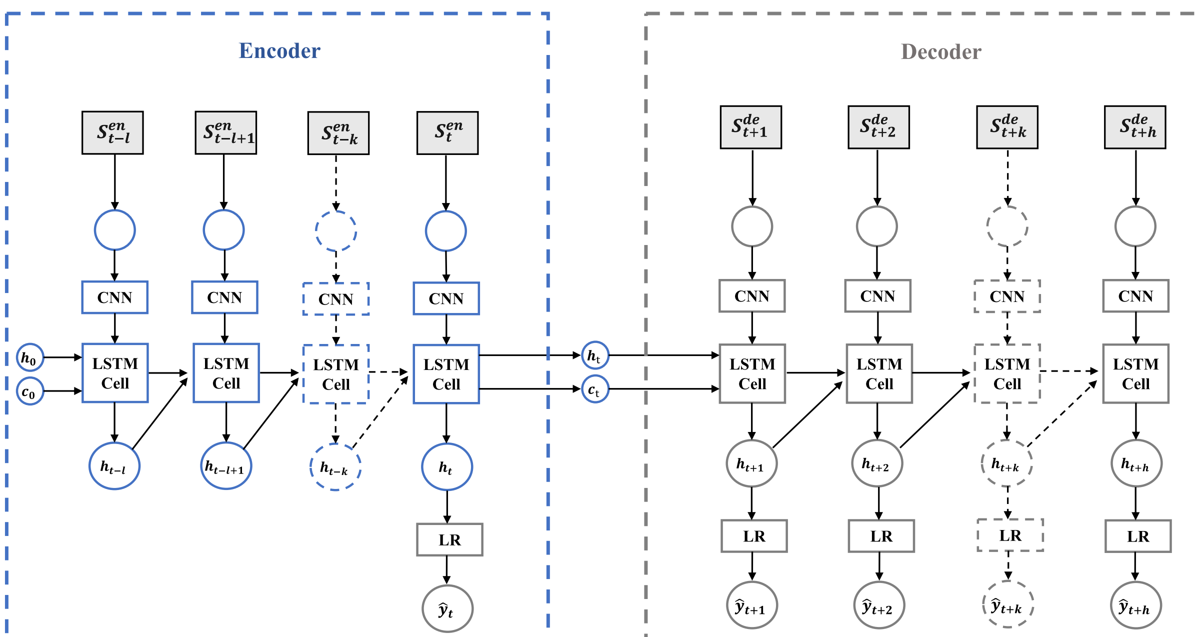


Fig. 6. The structure of EDC. The encoder and decoder are both CLSTM.

Table 5 shows the influence of input factors on the correction results. The publicly available dataset (solar plant 1) includes GHI and weather factors (AT and RH), while the actual field dataset (solar plant 2) only includes GHI and AT. The performance of Hybrid (GHI-AT) is better than that of Hybrid (GHI) in both solar plant 1 and solar plant 2, which shows that adding AT factor has a positive effect on improving the correction results. Hybrid (GHI-AT-RH) performs better than Hybrid (GHI-AT) which shows that using multi-dimensional data (GHI-AT-RH) as the input of DC can further help to improve the correction performance compared with using multi-dimensional data (GHI-AT) for solar plant 1. Therefore, the input of DC for solar plant 1 and solar plant 2 are multi-dimensional data (GHI-AT-RH) and multi-dimensional data (GHI-AT). Here, to distinguish the input factors, the proposed method Hybrid is marked as Hybrid (GHI-AT-RH) for solar plant 1 and Hybrid (GHI-AT) for solar plant 2. In the subsequent experiments, the proposed method is marked back to Hybrid.

4.2. Evaluation of different clustering methods

To verify the superiority of DC, this section compares DC with several different clustering algorithms, which are the K-means clustering algorithm, the K-means++ clustering algorithm and the Fuzzy C-Means (FCM) clustering algorithm. Combine these clustering algorithms with EDC respectively and compare the correction performance.

Table 6 shows the method Hybrid has the best performance among VMD-EDC combined with K-means clustering method, VMD-EDC combined with K-means++ clustering method, VMD-EDC combined with FCM clustering method (the results in the bold black mean the best performance). The above traditional clustering methods might ignore the features of multi-dimensional time series data when measuring distance between samples. DC uses the deep learning to map the original multi-dimensional information data into the Euclidean feature space and distance between samples is measured in the feature space, which can improve preciseness of clustering results. We also analyze the clustering results of DC using the Rand Index (RI) which is one of

Table 5
The day-ahead NWP GHI correction performance with different input factors.

Station	Models	Evaluation criteria		
		RMSE (W/m^2)	MAE (W/m^2)	nRMSE (%)
Solar plant 1	Hybrid(GHI)	82.75	38.10	32.25
	Hybrid(GHI-AT)	79.19	37.85	30.86
	Hybrid(GHI-AT-RH)	**75.51**	**34.21**	**29.43**
Solar plant 2	Hybrid(GHI)	131.33	67.25	58.60
	Hybrid(GHI-AT)	**120.64**	**62.56**	**53.83**

Table 6
The day-ahead NWP GHI correction performance of VMD-EDC combined with clustering algorithms.

Station	Models	Evaluation criteria		
		RMSE (W/m^2)	MAE (W/m^2)	nRMSE (%)
Solar plant 1	VMD-EDC	84.53	38.15	32.94
	VMD-EDC+K-means	86.78	38.40	33.82
	VMD-EDC+K-means++	83.87	37.28	32.69
	VMD-EDC+FCM	81.44	36.55	31.74
	Hybrid	**75.51**	**34.21**	**29.43**
Solar plant 2	VMD-EDC	138.36	72.78	61.74
	VMD-EDC+K-means	137.08	72.06	61.17
	VMD-EDC+K-means++	133.54	66.51	59.59
	VMD-EDC+FCM	131.01	64.15	58.46
	Hybrid	**120.64**	**62.56**	**53.83**

Table 7
Day-ahead NWP GHI correction performance comparisons of various models.

Station	Models	Evaluation criteria				
		RMSE (W/m^2)	MAE (W/m^2)	nRMSE (%)	nMAE (%)	MAPE (%)
Solar plant 1	NWP	109.08	50.47	42.51	19.67	26.22
	BPNN	97.05	41.53	37.82	16.19	24.51
	XGBoost	94.53	36.52	36.84	14.23	**23.26**
	SVR	101.22	37.42	39.45	14.58	23.93
	LSTM	96.77	37.02	37.71	14.43	25.25
	TFT	99.91	39.10	38.94	15.24	24.31
	EDC	90.75	**35.40**	35.36	**13.79**	23.80
	VMD-EDC+K-means++	**83.87**	37.28	**32.69**	14.53	23.62
	Hybrid	**75.71**	**34.21**	**29.43**	**13.33**	**22.33**
Solar plant 2	NWP	208.97	111.08	93.24	49.56	181.90
	BPNN	185.55	98.88	82.79	44.12	143.97
	XGBoost	165.75	85.17	73.96	38.01	107.16
	SVR	171.17	96.43	76.38	43.03	167.61
	LSTM	183.86	96.42	80.50	41.95	130.93
	TFT	170.39	91.50	76.03	40.83	104.79
	EDC	157.83	79.31	70.42	35.39	85.39
	VMD-EDC+K-means++	**133.54**	**66.51**	**59.59**	**29.68**	**78.12**
	Hybrid	**120.64**	**62.56**	**53.83**	**27.91**	**64.30**

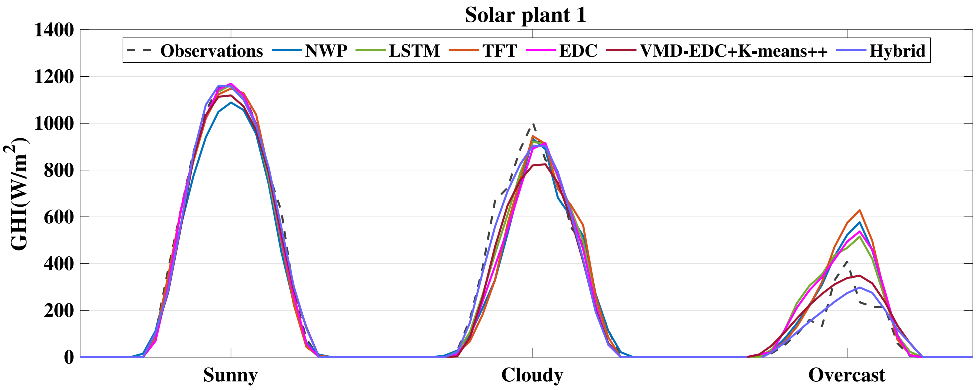
the measurements for evaluating the accuracy of the clustering results. The value range of RI is $[0, 1]$, and the larger the RI value is, the more consistent the clustering results are with the actual situation. For solar plant 1 and solar plant 2, the RI are 0.78 and 0.72, which indicates that the proposed DC can obtain clustering results relatively consistent with the actual situation.

4.3. Correction performance analysis

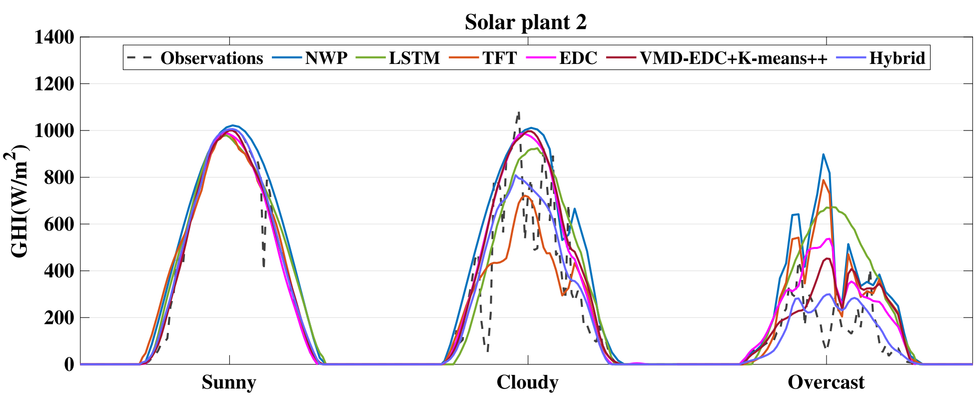
Various comparative experiments are performed in this subsection. The models for comparisons are BPNN, XGBoost, SVR, LSTM, TFT, EDC and VMD-EDC+K-means++. The RMSE, MAE, nRMSE, nMAE and MAPE are selected as evaluation criteria. It should be noted that for providing a fair comparison, all models adopt the same datasets. Table 7 shows the day-ahead NWP GHI correction results of different models with different evaluation criteria. The results in the bold black mean the best and second-best performance.

In terms of RMSE and nRMSE, two methods (the proposed method Hybrid and the method VMD-EDC+K-means++) containing the clustering part achieve the better performance in both solar plant 1 and solar plant 2. Taking solar plant 2 as an example, the Hybrid yields the smallest RMSE value (120.64 W/m^2) and the VMD-EDC+K-means++ yields the second-smallest RMSE value (133.54 W/m^2). Through the clustering methods, the information data of historical samples are categorized into several clusters and regression models are developed for each cluster, which is beneficial to improve the accuracy of correction results. More importantly, the hybrid performs more robust than VMD-EDC+K-means++ under different metrics, which may be due to the fact that adopting DC can avoid some sub-optimal clustering results. Also, EDC performs better than TFT in this work. Compared EDC with BPNN and SVR, the performance of EDC is obviously better, benefiting from an encoder–decoder framework. The EDC achieves a better performance than LSTM, which yield the MAPE with 85.39% and 130.93% respectively. Another reason is that CLSTM is applied to learning effective features of multi-dimensional data input, which can lead to better performance.

Real GHI series (black dotted lines), NWP GHI series (blue lines) and corrected GHI series from various models for different weather conditions are illustrated in Fig. 7. The correction accuracy of clustering-based methods is generally better under different weather conditions, indicating the effectiveness of clustering part. As shown in Figs. 7(a) and 7(b), most models can achieve relatively good correction results



(a)



(b)

Fig. 7. The GHI correction results of different models for sunny, cloudy, overcast weather condition in (a) Solar plant 1 and (b) Solar plant 2.

under sunny conditions. The Hybrid for solar plant 2 yields the smallest MAE value with 17.08 W/m^2. However, these correction models cannot be sensitive to sudden changes of GHI. For cloudy conditions, Fig. 7(b) shows that the original NWP GHI has relatively large error with the biggest MAE with 107.30 W/m^2 and the Hybrid achieves the best performance with the smallest MAE with 69.54 W/m^2. The observed GHI values under overcast conditions are generally low and the NWP GHI values are relatively large, resulting in a large NWP GHI error. The correction result curves of two hybrid methods with clustering algorithm are closer to the observed GHI curve. Overall, it is more robust for the Hybrid to achieve better performance under different weather conditions compared with the above methods.

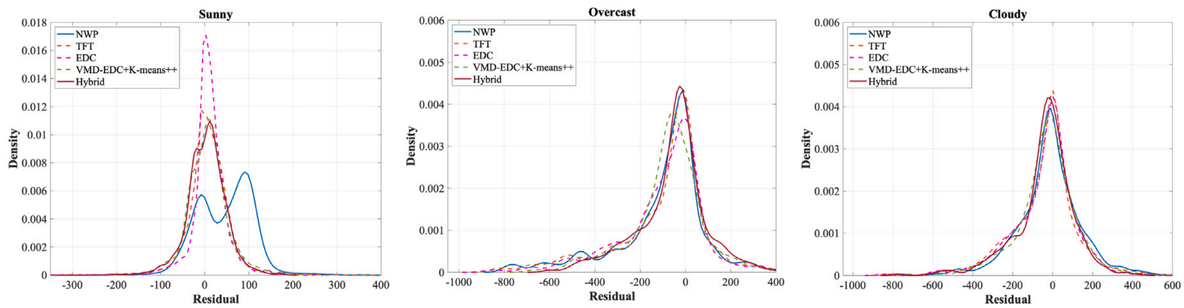
Error density analysis is shown in Fig. 8 to further view the distribution of masked errors under three weather conditions [38]. In most cases, the error distribution of the proposed method is with a high density of residuals close to 0 and a low density of higher residuals (tails of the distribution). In particular, our proposed method has obvious superiority under all three weather conditions. EDC as a part of our proposed Hybrid method has a good performance under sunny and cloudy weather conditions (especially in solar plant (1), while EDC does not work well under overcast condition, illustrating that the robustness of the NWP GHI correction decreases without introducing DC.

The correlation distribution between corrected GHI values and actual GHI values where observed GHI values are over 400 W/m^2 is displayed in Fig. 9. The reason for choosing more than 400 W/m^2 is that the comparison of the performance of different models is more concerned when the GHI is higher [39]. The orange dots and blue dots denote NWP GHI values and the corrected GHI values obtained by the Hybrid respectively. Red lines in the plots denote the actual observed GHI values and the forecast GHI values are the same.

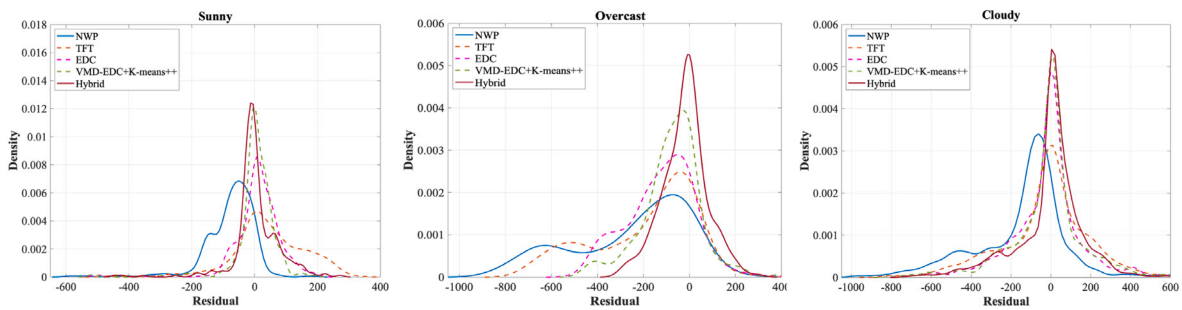
As shown in Fig. 9, for solar plant 1, GHI forecasts obtained by NWP have an obvious negative bias with GHI observations. The corrected GHI forecasts obtained by Hybrid are more accurate although it also introduces a relatively small negative bias. For solar plant 2, GHI forecasts obtained by NWP have a positive bias with GHI observations while the corrected GHI forecasts obtained by Hybrid have a negative bias. When the observed GHI values are over 400 W/m^2, the GHI forecast values obtained by the correction model are generally lower than the observed values. This may be because the model cannot precisely forecast the dynamic characteristics of GHI. The temporal resolution of solar plant 1 and solar plant 2 is 1 h and 15 min respectively, and the dynamic characteristics of GHI in solar plant 2 are more obvious. Therefore, when the observed GHI values are over 400 W/m^2, the correction results in solar plant 2 have a more obvious negative bias compared with solar plant 1. This is a dilemma to improve the robustness of the model to the dynamic characteristics of GHI. Compared with NWP GHI values, the correction results obtained by the Hybrid are more consistent with the actual GHI observations. However, there is still some big error compared with the actual value. It may be because the original error of NWP information data in the input of DC is too large, which leads to wrong cluster assignment of DC. Research shall further consider enhancing clustering effect by adjusting input data components.

5. Conclusions

To improve the day-ahead GHI forecast accuracy of NWP, the NWP GHI error analysis is first carried out and the results show that the NWP GHI error characteristic varies significantly under different

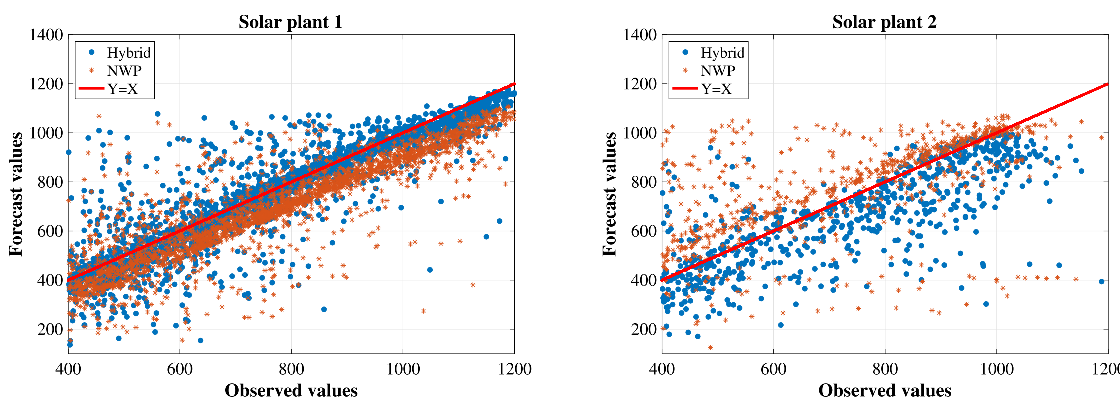


(a) Solar plant 1



(b) Solar plant 2

Fig. 8. Error distribution.



(a) (b)

Fig. 9. The scatter plots of the observed and corrected GHI (over 400 W/m^2) for the hybrid method in (a) Solar plant 1 and (b) Solar plant 2.

weather conditions. This means that it should be modeled separately for different weather conditions. The present study develops a hybrid correction model based on deep learning to correct NWP GHI for different weather conditions. The DC is applied to categorize the information series of historical samples into several clusters. The advantage of DC is that a more accurate clustering performance can be provided, benefitting from the simultaneous optimization of feature learning and clustering assignment. The observed GHI series is decomposed into several sub-series by VMD algorithm for feature exploitation. Then, EDC is trained for day-ahead NWP GHI correction of each cluster. The CLSTM is utilized to learn effective features of multi-dimensional input data in both the clustering step and regression step. The performance of the proposed model is evaluated by five statistical parameters, namely, RMSE, MAE, nRMSE, nMAE and MAPE. The effectiveness of the proposed hybrid method has been demonstrated by experiments with both publicly available dataset and field measurements of an actual photovoltaic (PV) power station. To verify the superiority of the proposed correction model, the performances of BPNN, XGBoost, SVR, LSTM, TFT, EDC and VMD-EDC+K-means++ are tested with the same datasets. The results have verified the outstanding ability of the proposed method to improve the correction accuracy of day-ahead GHI forecast for NWP. Moreover, the proposed correction model allows a more accurate forecast of PV power generation.

According to above performance analysis, it can be noticed that the NWP GHI values can be well corrected by the proposed Hybrid. However, it requires a large amount of computation in training different models for each cluster. The proposed method is data-driven and the historical samples with similar patterns are divided into the same cluster as accurately as possible by DC. It is inevitable that there may be some sub-optimal clustering results or misclassification results, which may affect the correction accuracy. Future work will study how to further eliminate the correction error caused by clustering bias. This work is based on a single PV power station and further research will be carried out in combination with multiple PV power stations.

CRediT authorship contribution statement

Weijing Dou: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Kai Wang:** Methodology, Formal analysis. **Shuo Shan:** Writing – review & editing. **Chenxi Li:** Writing – review & editing. **Yiye Wang:** Writing – review & editing. **Kanjian Zhang:** Project administration, Funding acquisition, Conceptualization. **Haikun Wei:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Victor Sreeram:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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